

Research Paper Writing Seminar I (RPW1)

Session 01: Introduction to RPW1

AY2026, S2 (Q3 and Q4)

RPW1 Team

University of Aizu

Agenda

- Course overview
- Research papers and why we write them
- The research and writing process
- Peer review and revision
- Q&A and recap

Our university motto

to Advance Knowledge for Humanity

Our founding principle calls us to **invent** and **discover** for the peace and prosperity of humankind.



My personal take

- We are here to invent new technologies, discover useful things, and create real value for our local community and the world.
- We are NOT here simply to publish as many papers as possible.
- **Papers are a way to communicate our discoveries — not the ultimate goal of our research.**

Teaching team

The AY2026 teaching team includes:

- KACHI Yasuyuki: Mathematics
- ROY Debopriyo: Linguistics
- VIGLIETTA Giovanni: Theory of Informatics
- NASSANI Alaeddin: Augmented Reality
- DANG Nam Khanh: Computer Architecture

We also have several invited lecturers!

RPW1 Survey



Research experience

89% had experience
11% had no experience



Paper writing

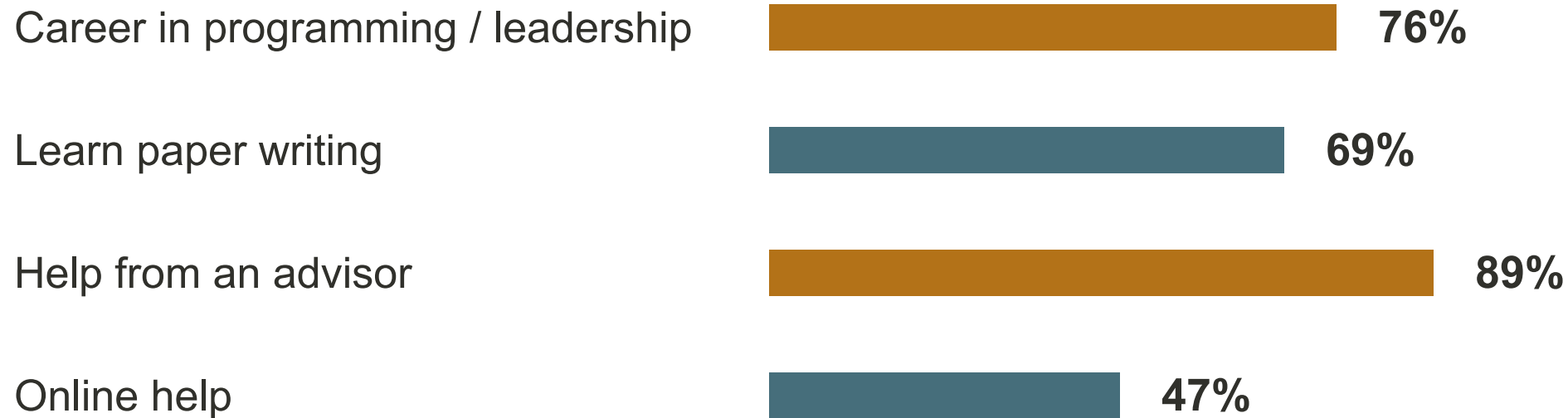
62% writing or had written
38% had never written

2025 survey: 45 responses · Writing difficulty: 7.22 / 10

These results describe the 2025 class.

RPW1 Survey: goals and support

2025 survey · 45 responses



Multiple answers were allowed

These results describe the 2025 class.

Common writing difficulties

- Explaining an idea clearly
- Organizing sections and connecting ideas
- Academic English and word choice
- Choosing a research topic
- Reviewing related work
- Using LaTeX

Course goals

What RPW1 covers

- Understand the structure of a research paper.
- Develop skills for research paper writing.
- Learn citation, submission, and ethical writing practices.

Course scope

- Your supervisor guides your research.
- RPW1 focuses on scientific writing. General English study is separate.

Research fields

The course uses examples from computer science and engineering.

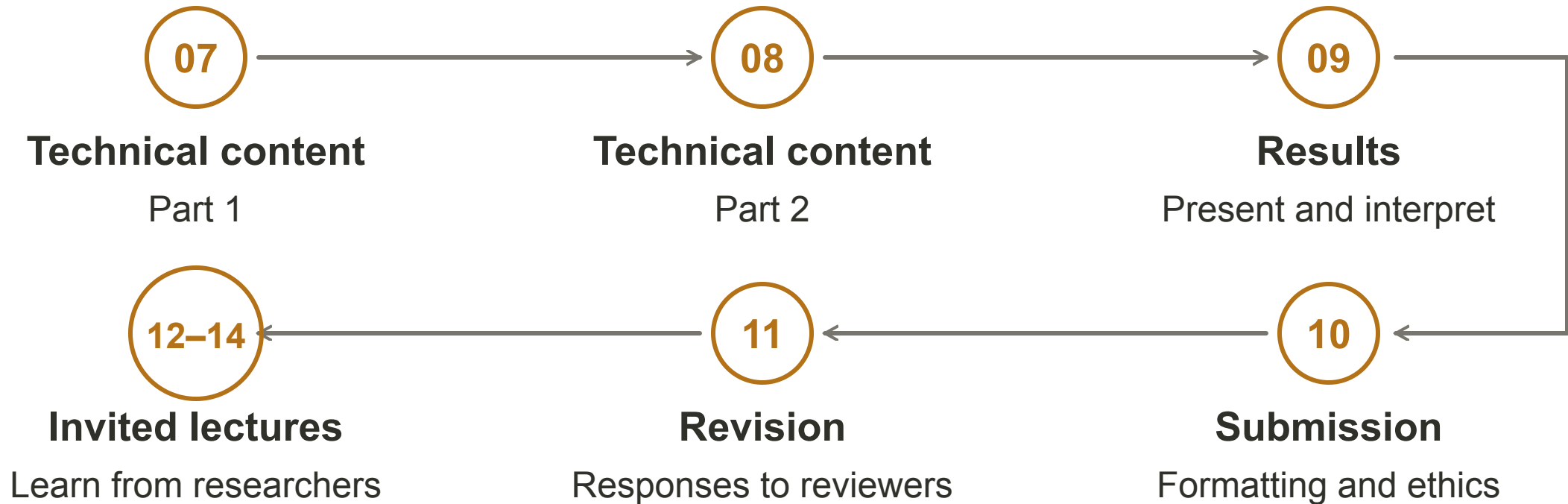
- Computer systems and networks
- Security, software, and databases
- Theory and algorithms
- AI, graphics, and multimedia
- Human–computer interaction
- Other areas of computer science and engineering

Course plan: Sessions 01–06



Check ELMS for the confirmed order and dates. Topics and dates may change.

Course plan: Sessions 07–14



Check ELMS for the confirmed order and dates. Topics and dates may change.

Course information in ELMS

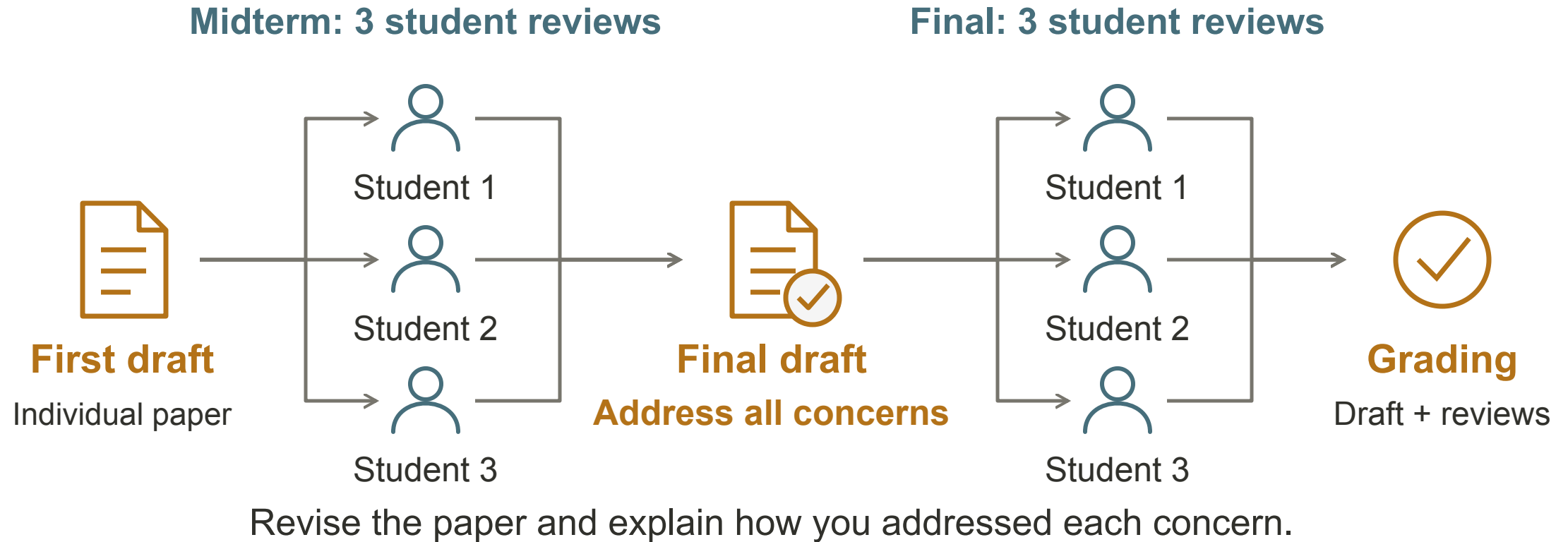
Use the **AY2026 RPW1 ELMS course page** for:

- Class dates, rooms, and schedule changes
- Attendance and grading rules
- Plagiarism and generative AI rules
- Slides, questionnaires, and assignments

Planned class time: periods 9–10, **17:00–18:40**, on Monday, Tuesday, or Wednesday.

Check ELMS and your email for the date of each class and any changes.

Midterm and final assignment



Your final draft must address concerns from all three midterm reviewers.

Individual work and course rules

The AY2026 rules are posted on ELMS:

- Individual authorship and permitted help
- Citation and reuse of research material
- Generative AI use and disclosure
- Submission format and review responsibilities

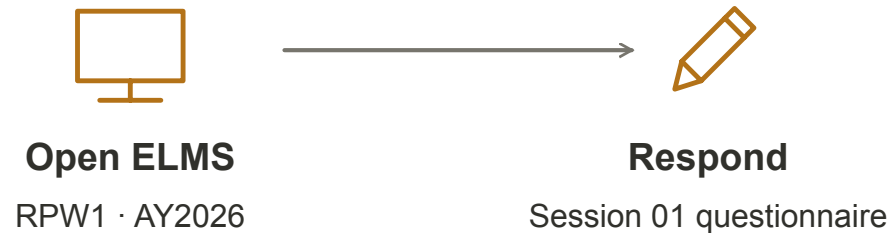
Ask the instructors if a rule is unclear.

Grading

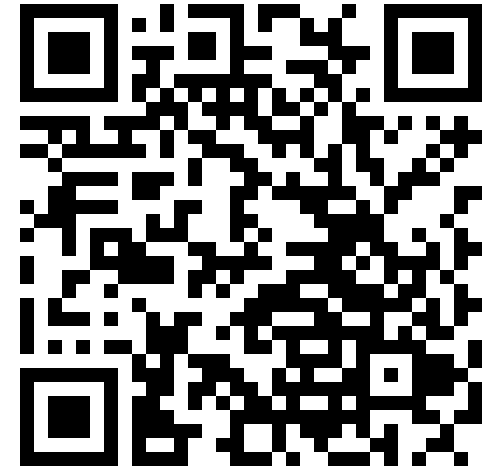
- **50%: Class activities**, including your peer reviews
- **50%: Paper draft**, with midterm and final rounds

Check ELMS for the detailed rubric and deadlines.

Session 01: Q&A and survey



- Open ELMS: <https://elms.u-aizu.ac.jp/>
- Please complete the questionnaire in 10 minutes.
- The survey closes at **17:50**.



What is a research paper?

Common types of research papers

PAPER TYPE		MAIN CONTRIBUTION
	Survey / review	Summarizes earlier studies and identifies open questions
	Finding / empirical study	Reports new observations or experimental results
	Solution / method	Proposes an approach and evaluates how well it works
	Theory	Develops models or proves new results
	Resource / dataset	Introduces data or tools for further research

What a research paper explains



- What problem the research addresses
- How the work relates to earlier studies
- What the authors did
- What evidence supports the conclusions
- What limits remain

Example: A systems paper

Pollux: Co-adaptive Cluster Scheduling for Goodput-Optimized Deep Learning

The first page includes a title, authors, abstract, and introduction.

How does the abstract explain the problem and result?

Pollux: Co-adaptive Cluster Scheduling for Goodput-Optimized Deep Learning

Aurick Qiao^{1,2} Sang Keun Choe² Suhas Jayaram Subramanya² Willie Neiswanger^{1,2}
Qirong Ho¹ Hao Zhang^{1,3} Gregory R. Ganger² Eric P. King^{4,1,2}

¹Petuum, Inc. ²Carnegie Mellon University ³UC Berkeley ⁴MBZUAI

Abstract

Pollux improves scheduling performance in deep learning (DL) clusters by adaptively co-optimizing inter-dependent factors both at the per-job level and at the cluster-wide level. Most existing schedulers expect users to specify the number of resources for each job, often leading to inefficient resource use. Some recent schedulers choose job resources for users, but do so without awareness of how DL training can be re-optimized to better utilize the provided resources.

Pollux simultaneously considers both aspects. By monitoring the status of each job during training, Pollux models how their *goodput* (a metric we introduce to combine system throughput with statistical efficiency) would change by adding or removing resources. Pollux dynamically (re-)assigns resources to improve cluster-wide goodput, while respecting fairness and continually optimizing each DL job to better utilize these resources.

In experiments with real DL jobs and with trace-driven simulations, Pollux reduces average job completion times by 37–50% relative to state-of-the-art DL schedulers, even when they are provided with ideal resource and training configurations for every job. Pollux promotes fairness among DL jobs competing for resources, based on a more meaningful measure of *weight* (job progress), and reveals a new opportunity for reducing DL cost in cloud environments. Pollux is implemented and publicly available as part of an open-source project at <https://github.com/petuum/adaptDL>.

1 Introduction

Deep learning (DL) training has rapidly become a dominant workload in many shared resource environments such as datacenters and the cloud. DL jobs are resource-intensive and long-running, often demanding distributed execution using expensive hardware devices (e.g. GPUs or TPUs) in order to complete within reasonable amounts of time. To meet this resource demand, dedicated clusters are often provisioned for deep learning [31, 67], with a scheduler that mediates resource sharing between many competing DL jobs.

Existing schedulers require users to manually configure their jobs, which if done improperly, can greatly degrade training performance and resource efficiency. For example, allocating too many GPUs may result in long queuing times and inefficient resource usage, while allocating too few GPUs may result in long runtimes and unused resources. Such decisions are especially difficult to make in a shared-cluster setting, since optimal choices are dynamic and depend on the cluster load while a job is running.

Even though recent elastic schedulers can automatically select an appropriate amount of resources for each job, they do so blindly to inter-dependent training-related configurations that are just as important. For example, the *batch size* and *learning rate* of a DL job influence the amount of computation needed to train its model. These optimal choices vary between different DL tasks and model architectures, and they have strong dependence on the job's allocation of resources.

The amount of resources, batch size, and learning rate are difficult to configure appropriately without expert knowledge about both the cluster hardware performance and DL model architecture. Due to the inter-dependence between their optimal values, they should be configured jointly with each other. Due to the dynamic nature of shared clusters, their optimal values may change over time. This creates a complex web of considerations a user must make in order to configure their job for efficient execution and resource utilization.

How can a cluster scheduler help to automatically configure user-submitted DL jobs? Fundamentally, a properly-configured DL job strikes a balance between two often opposing desires: (1) *system throughput*, the number of training examples processed per wall-clock time, and (2) *statistical efficiency*, the amount of progress made per training example processed.

System throughput can be increased by increasing the batch size, as illustrated in Fig. 1a. A larger batch size enables higher utilization of more compute resources (e.g., more GPUs). But, even with an optimally-tuned learning rate, increasing the batch size often results in a decreased statistical efficiency [46, 57]. For every distinct allocation of GPUs, there is potentially a different batch size that best balances increasing

Example: A machine learning paper

Attention Is All You Need

Look at the title, author list, abstract, and section headings.

How do they help you find the main idea?

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.9 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

1 Introduction

Recurrent neural networks, long short-term memory [12] and gated recurrent [7] neural networks in particular, have been firmly established as state-of-the-art approaches in sequence modeling and transduction problems such as language modeling and machine translation [29, 21, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [31, 21, 11].

^{*}Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

[†]Work performed while at Google Brain.
[‡]Work performed while at Google Research.

31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

Why RPW1?

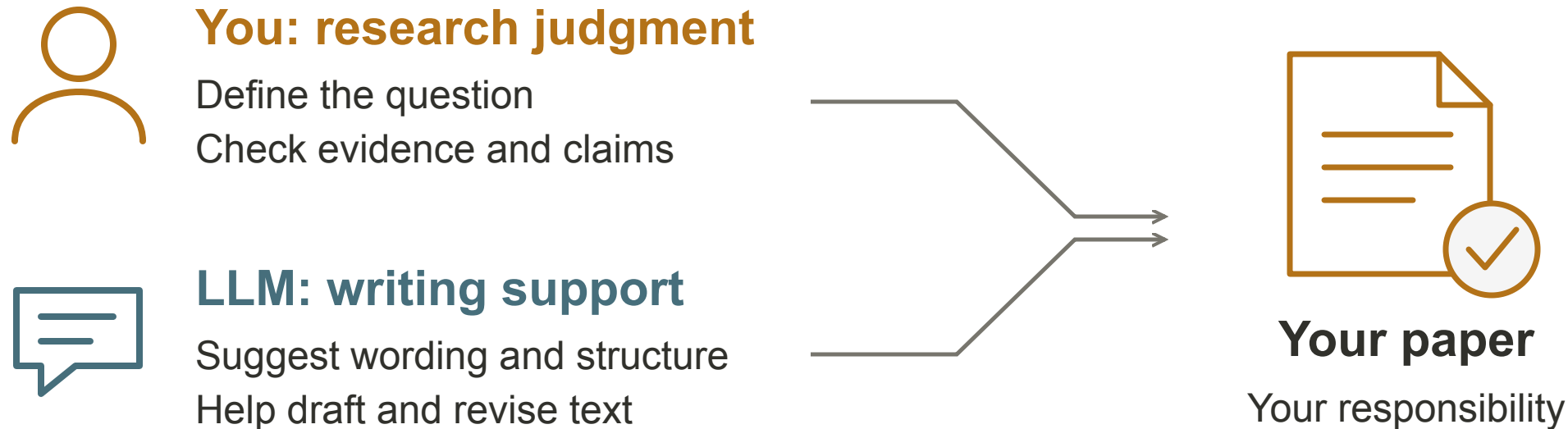
Why research writing matters



Writing makes your reasoning visible.

Why RPW1 in the era of LLMs?

- Large language models (LLMs) can help draft and revise text.



- You need to build and check a clear research argument.

Benefits of research writing

Academic:

- Develops analytical thinking and study design skills.
- Deepens understanding of the research topic.
- Improves how you explain methods and results.

Professional:

- Helps you communicate your expertise.
- Supports preparation for academic and industrial R&D roles.

Personal:

- Encourages intellectual curiosity.

Skills you can use beyond RPW1

- Write a research paper
- Explain your work in a job interview
- Describe an invention in a patent application
- Write a funding proposal
- Pitch an idea for a startup

Example: LLM training costs

Manager: "Training our large language model costs too much."

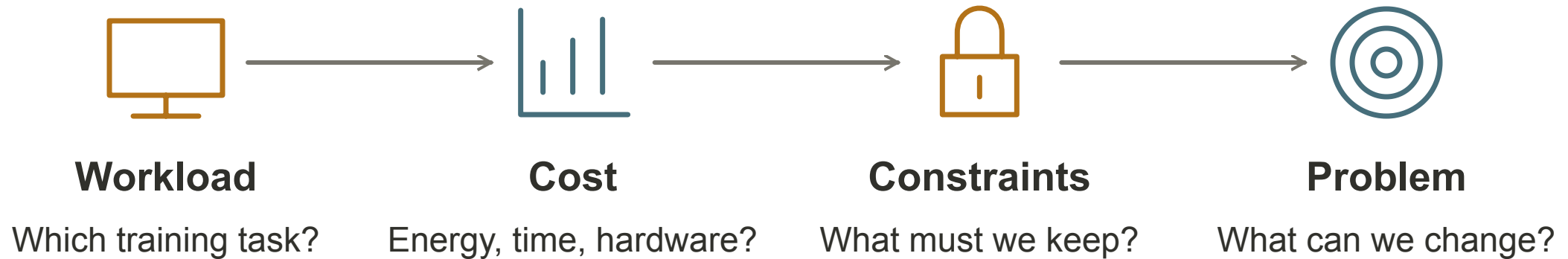
Possible replies:

- "Buy cheaper GPUs."
- "Rent cheaper computing resources."
- "Measure the costs first."
- "Reduce our salaries!"

Which reply helps us understand the problem?



Defining the research problem



The problem must be **clearly defined**.

Evidence from earlier work



Read the study

Hardware and workload

Measurement method

Reported limits



Check the claim

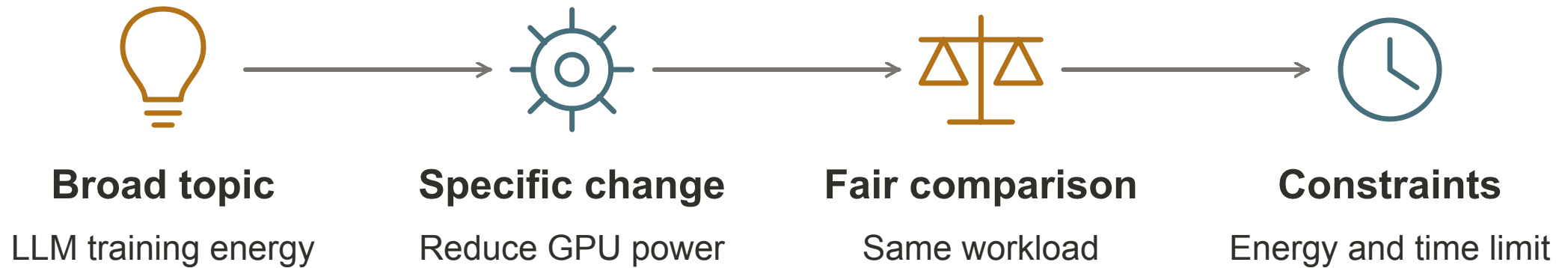
Training or inference?

Does it fit our question?

Can we trace the source?

A trusted source still needs **careful reading**.

A focused research question



Example research question:

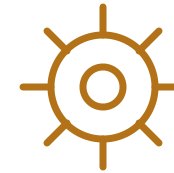
- **Can GPU clock settings reduce training energy within a time limit?**

A method and a fair test



Baseline

Same workload
Same hardware

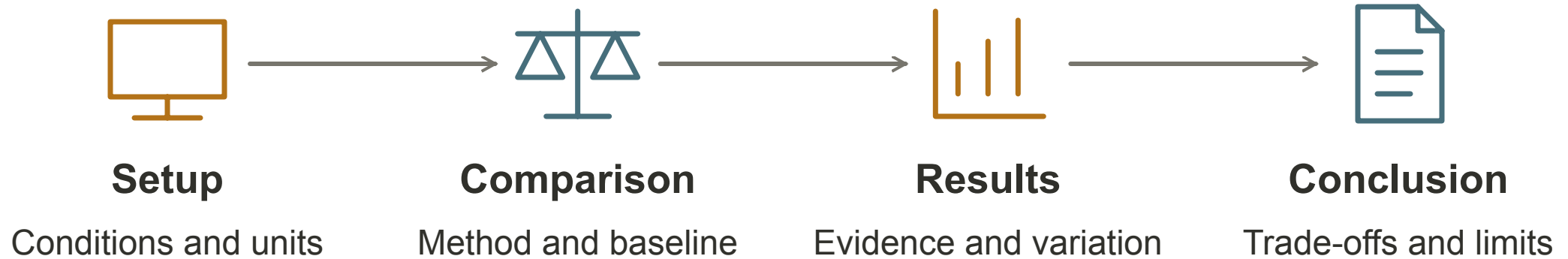


Proposed method

Same workload
Same hardware

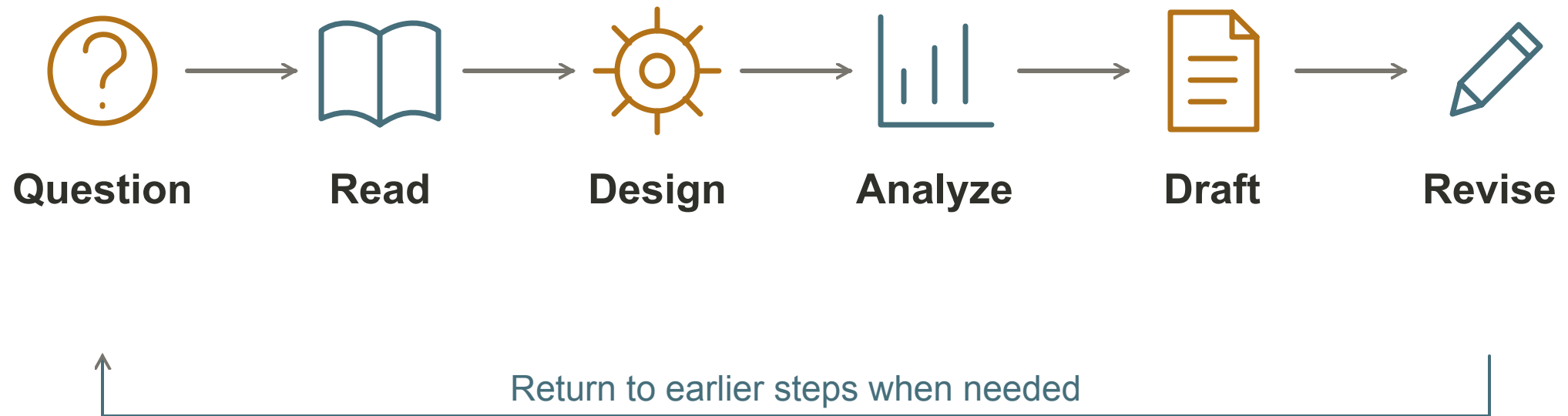
Compare energy, training time, and model quality. Repeat tests.

Reporting the result



The research and writing process

The research and writing process



A useful research question

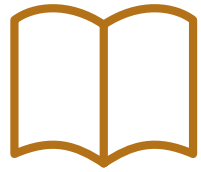
A useful question is:

- Clear enough for readers to understand
- Narrow enough for the planned study
- Possible to answer with available resources
- Important to the research community

Example:

- Weak: How does AI improve healthcare?
- Better: How does image augmentation affect a fixed model's COVID-19 classification accuracy on a held-out chest X-ray test set?

Literature review



Read

Problem and method



Compare

Results and limits



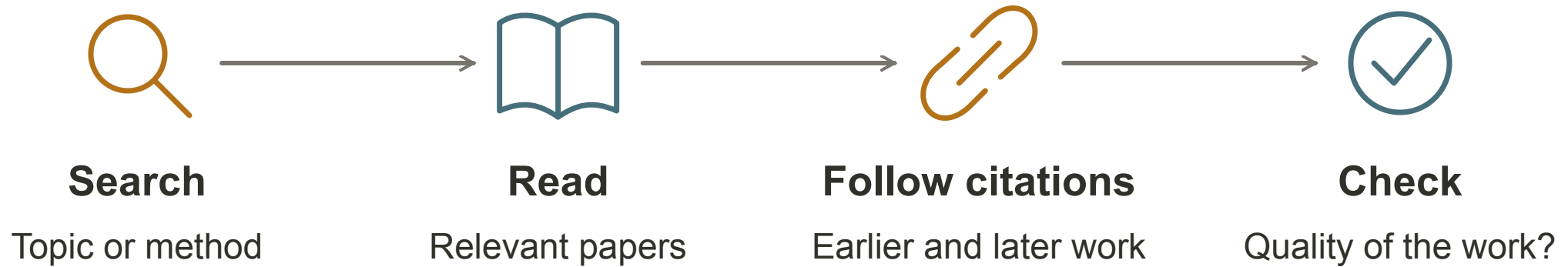
Connect

Relation to your study

- Explain how earlier work relates to your question.
- Identify existing research gaps.
- Support the development of your methodology.

We will cover this in Session 02.

Finding relevant papers



[Google Scholar](#) · [IEEE Xplore](#) · [ACM DL](#) · [arXiv](#)

AI search tools can help. Verify papers and references against the original sources.

We will cover this in Session 02.

Paper structure



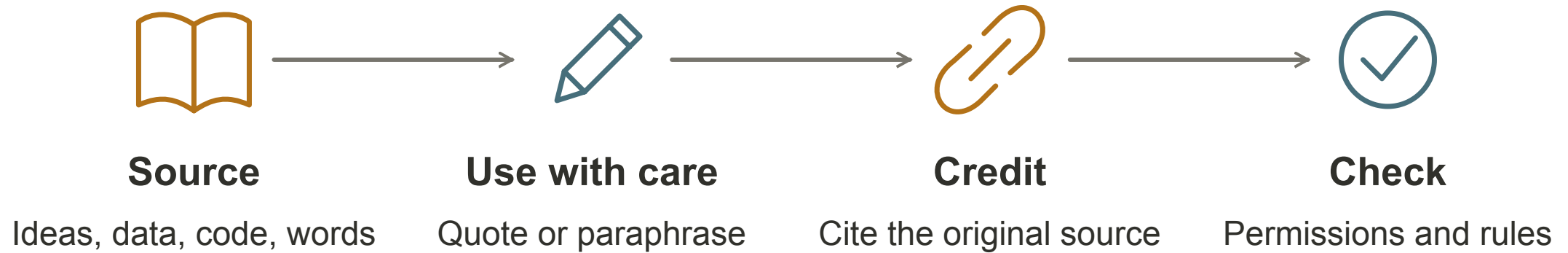
Section names and order vary by field and venue.

Paper structure (continued)



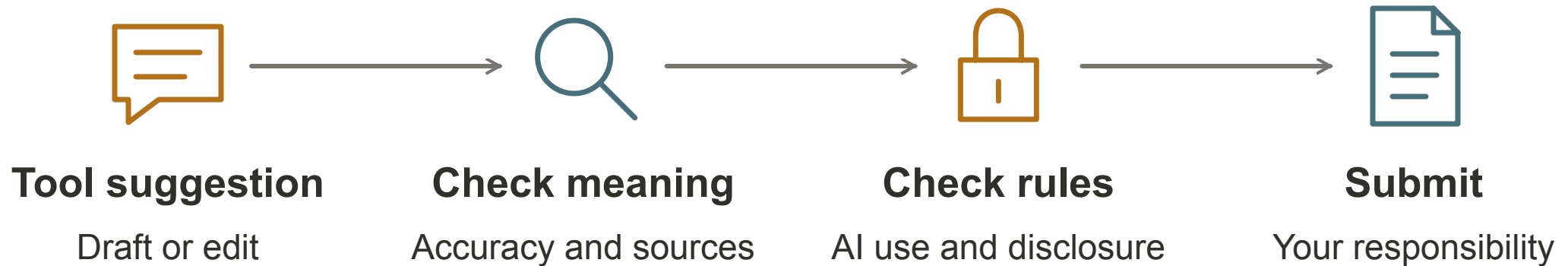
Some papers combine sections or use different names.

Citations and plagiarism



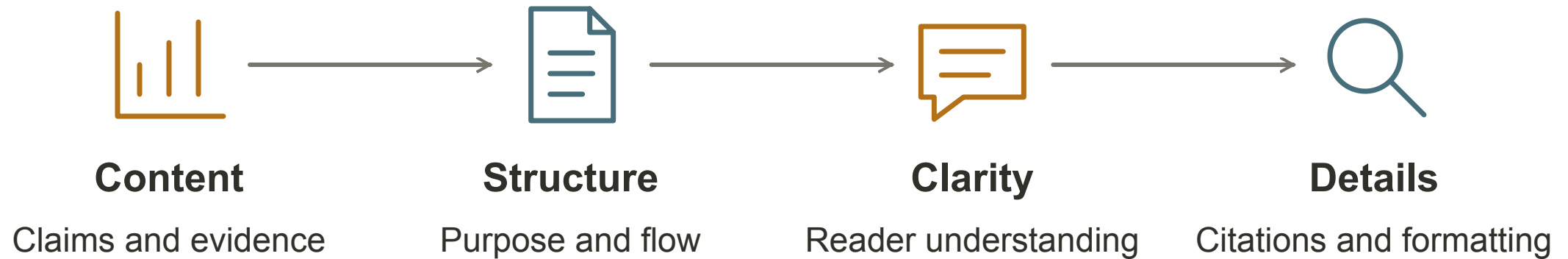
- Cite your own previous work when you reuse or extend it.
- Be cautious of AI hallucinations, and verify AI-generated claims and references.
- Check permissions before reproducing figures, tables, or text.

Writing tools and generative AI



You remain responsible for the work you submit.

Revision



Ask a reader to explain your main point back to you.

Common problems in a draft

- A broad problem with no clear research question
- A contribution that readers cannot identify
- Claims without enough evidence
- A list of papers without comparison or analysis
- Figures or explanations that are hard to follow
- Missing citations or incorrect formatting

Fix the main argument before polishing individual sentences.

Writing and reference tools

Task	Tools
Writing and formatting	Word, LaTeX, Overleaf
Managing references	Zotero, JabRef, Mendeley
Drafting and revision support	LLMs, where permitted
Similarity checking	Turnitin, iThenticate
Grammar check	Grammarly

Choose tools that fit your supervisor's workflow and the venue.
Check the output yourself.

How will your papers be reviewed?

Peer review

Peer review asks other researchers to assess a paper's quality and validity.

1. Initial screening: editorial office and editor / chair

- The office checks format and required information.
- The editor or chair checks scope and suitability for review.
- The paper may be returned for corrections or **rejected before review**.

2. Handling editor / area or track chair

- Assigns reviewers and manages the review process.
- Considers reports and recommends or makes a decision.

Peer review (continued)

3. Reviewers

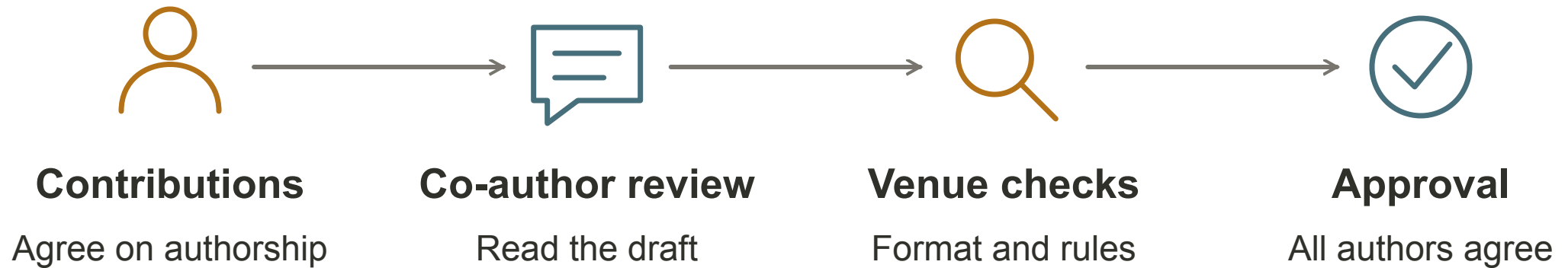
- Are invited to evaluate the paper.
- Assess its novelty, technical quality, clarity, and significance.
- Are usually anonymous, depending on the venue's review policy.

4. Decision authority: editor or program committee leadership

- Considers the paper, review reports, and recommendations.
- Makes or approves the **final decision**, under the venue's policy.

Roles and decision authority vary by venue.

Before submission



Submission and review



The number of reviewers and steps depend on the venue.

Anonymous review and conflicts



Single-anonymous

Reviewer identities hidden

Author identities visible



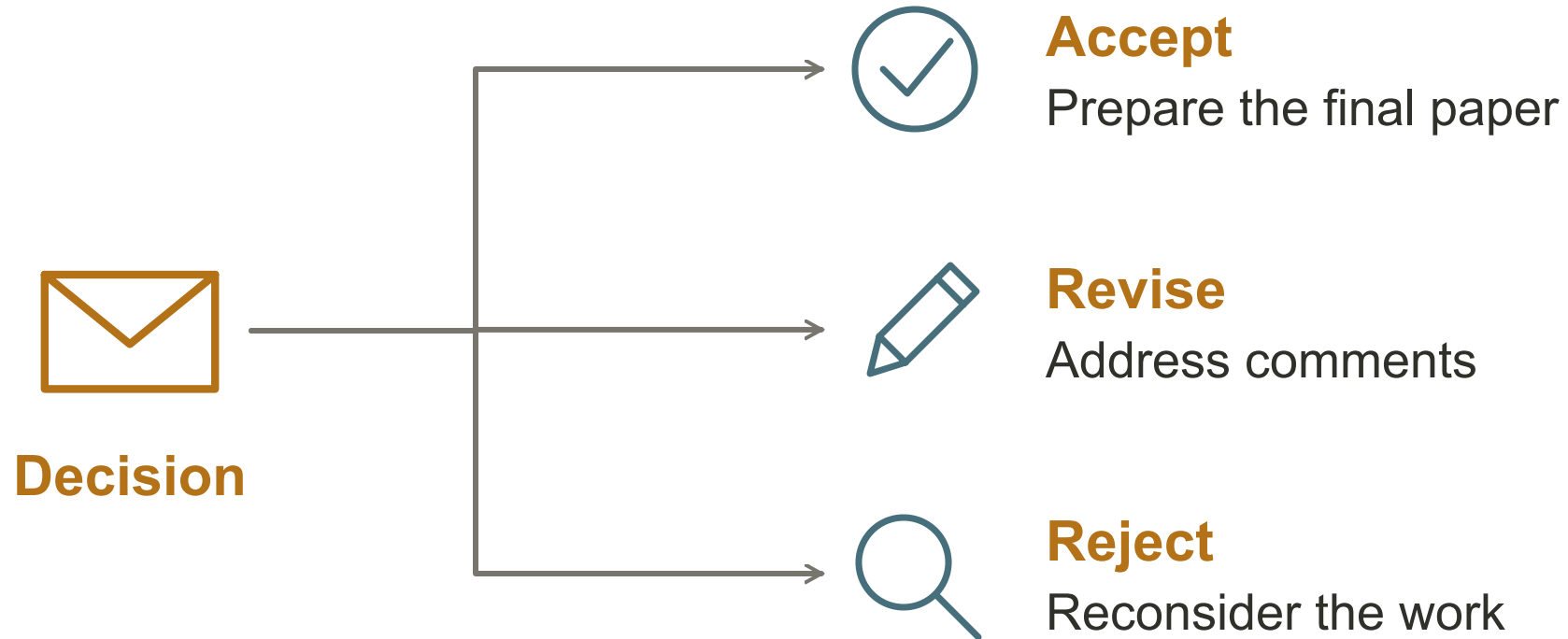
Double-anonymous

Reviewer identities hidden

Author identities hidden

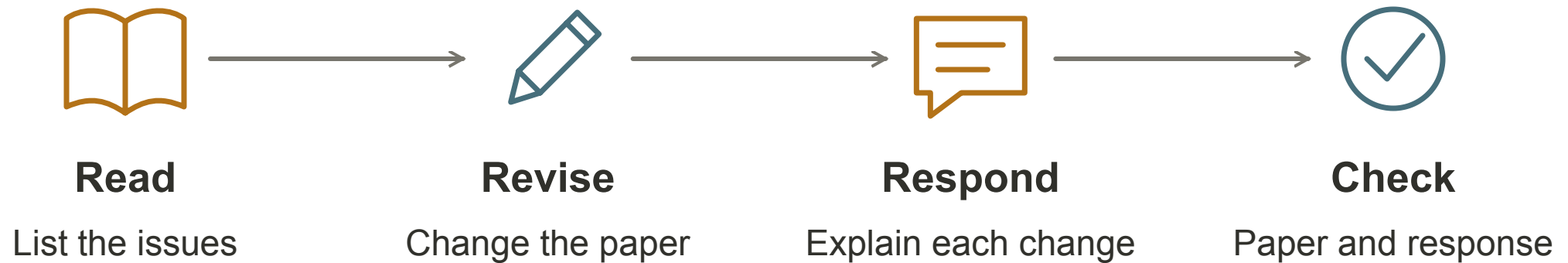
Reviewers need relevant expertise and must disclose conflicts.

Possible decisions



A request for revision or an author response does not guarantee acceptance.

Responding to reviews



Support disagreements with reasons and evidence.

A useful review



Vague feedback

“The experiments
are weak.”



Useful feedback

“Test a second workload
to show whether the result
applies more widely.”

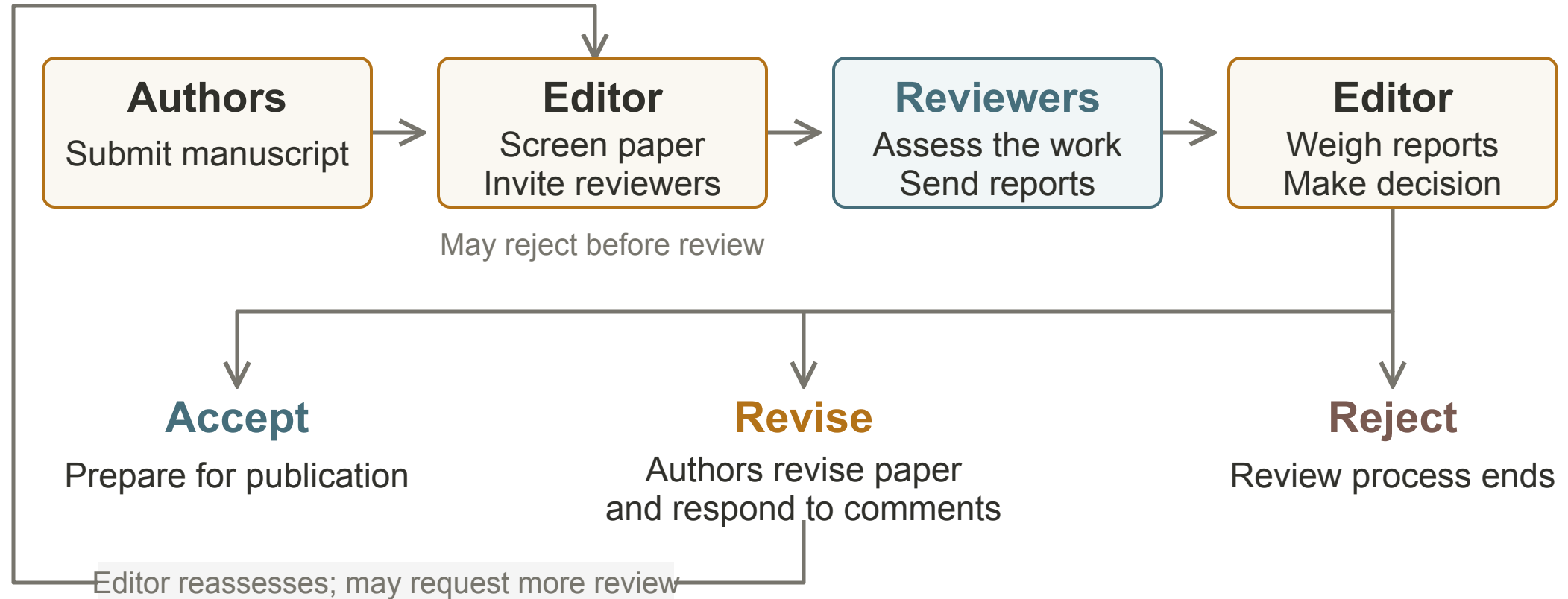
Be specific, fair, and constructive. Keep reviews confidential.

Reviewers and decision makers

- Reviewers are usually experts in the specific research field.
- Editors and program committee chairs
 - Have broad knowledge of the research area.
 - May not be experts in every specific topic.
 - Usually give significant consideration to the reviewers' recommendations when making the final decision.

Revision and rebuttal: we will cover this in Session 11.

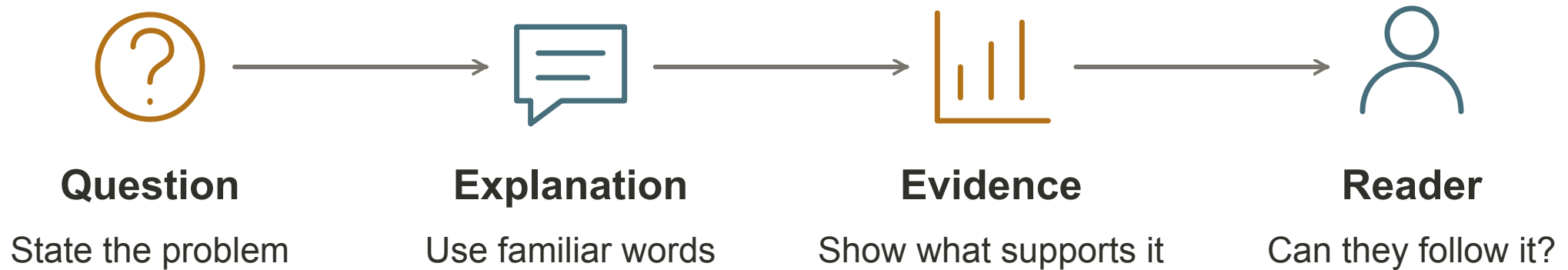
Peer review: overall flow



Typical journal process. Conference review may have fewer revision rounds.

Some modest suggestions

Clear scientific writing



- Precision matters more than impressive language.
- Clear explanations help busy editors and reviewers assess your work.
- Readers need to understand the contribution and its supporting evidence.

A precise sentence



Vague claim

“Our method
is much better.”



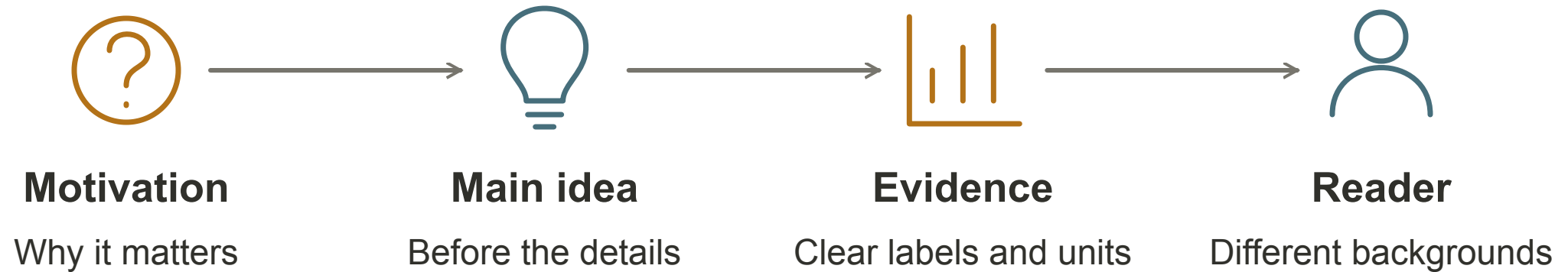
Precise claim

“In this test, our method
used 10% less energy
than the baseline.”

Hypothetical writing example, not a research result.

Explain the workload, measurement method, and variation.

Writing for your readers



Conferences and journals



Conference

Often a fixed deadline
Scheduled event
Event-based review stages



Journal

Often year-round submission
Published in a journal
May have revision rounds

Page limits, review time, and procedures vary.

Read the current venue instructions.

Choosing a publication venue

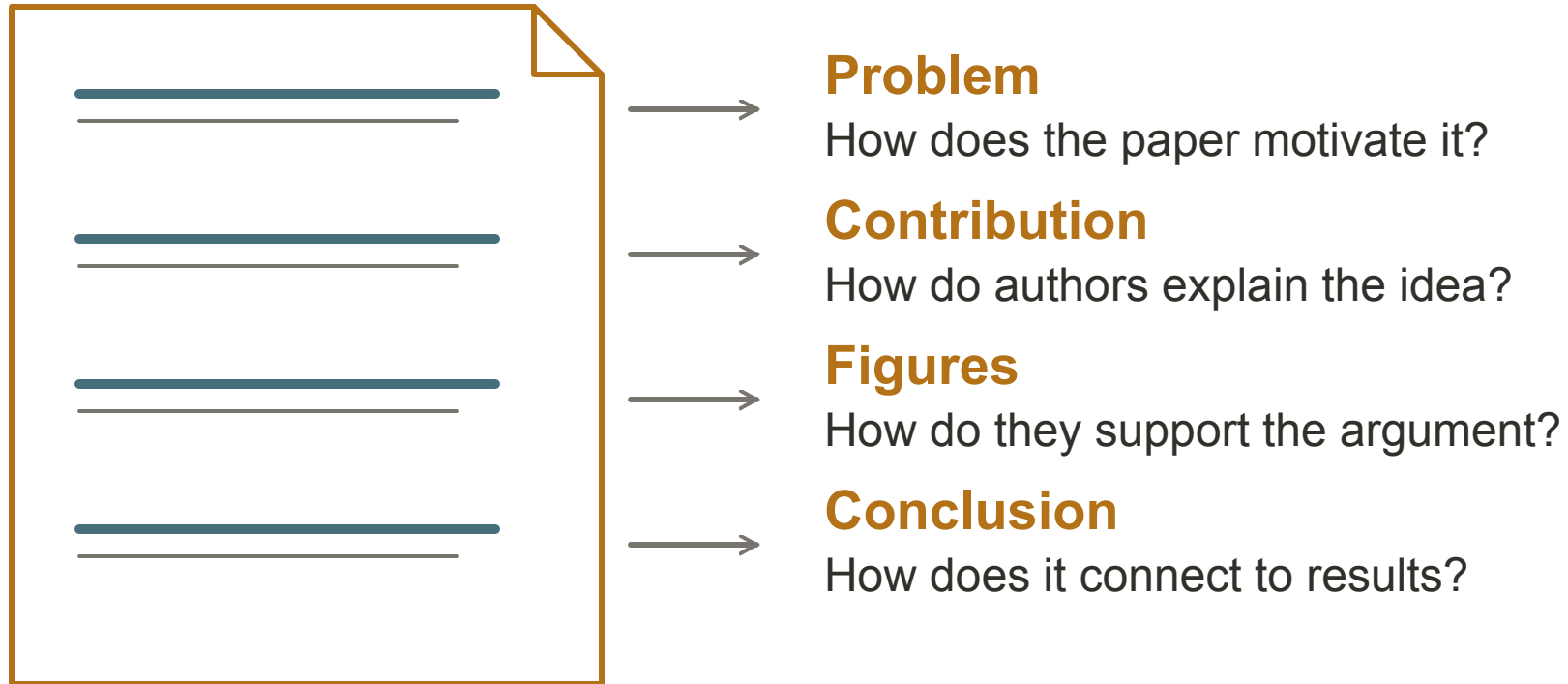
Discuss the choice with your supervisor.

- Does the venue publish work on your topic?
- Who is the intended audience?
- Are the review process and publisher clear?
- Can you meet the format and deadline?
- Are publication costs and policies acceptable?

Use rankings as one source of information.

Read recent papers and assess their relevance.

Learning from published papers



Learn from the structure. Write your own explanation.

Venue rankings and citation metrics

- Computing venue lists: [ICORE \(formerly CORE\)](#) and CCF
- Journal indicators: SCImago Journal Rank and Journal Impact Factor
- Publication citation metrics: [Google Scholar Metrics](#)

These indicators provide context. They do not establish the quality of an individual paper.

Read the papers and discuss venue choices with your supervisor.

Q&A time

Questions and review



Q&A: What would you like to ask about the course?

Four points to remember

- Be precise.
- Use simple language.
- Improve the paper through revision.
- Focus on quality rather than quantity.



to Advance Knowledge for Humanity

Our founding principle calls us to **invent**
and **discover** for the peace and
prosperity of humankind.



Thank you

Next session: Literature survey and review